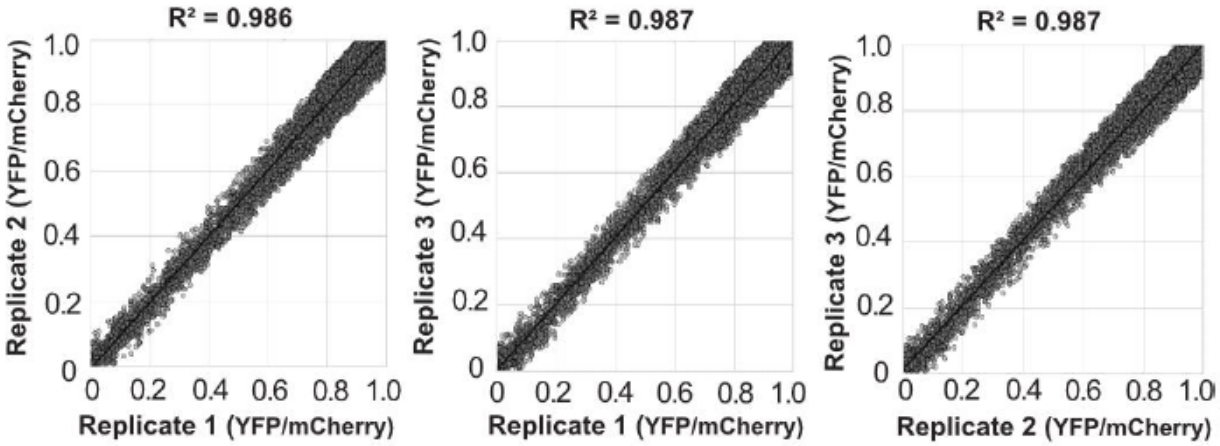


FIGURE S1 | Reproducibility of FACS-Seq measurements of mean YFP levels for 5' TLs. The scatter plots show the comparison of three replicates. R^2 values are listed above each plot. Note that reporters with noisy expression measurements (coefficient of variation > 0.05) were removed from this and other analyses in this study, as previously reported (34).

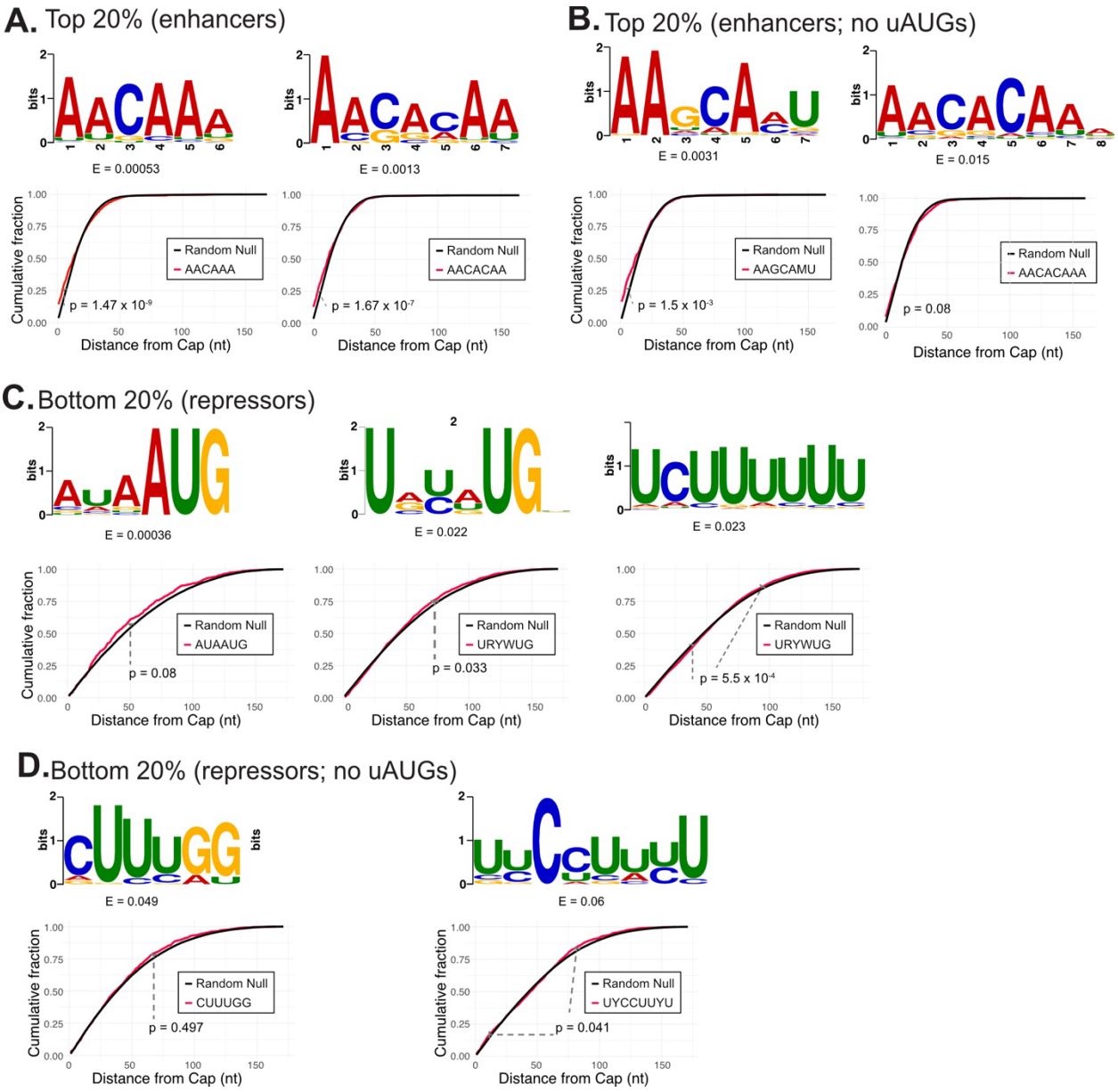


FIGURE S2. Positional biases of motifs associated with high and low RNA levels. Motifs were identified using the STREME algorithm from the MEME suite (c.f. Figure 1). **A.** Sequences in transcript leaders from the top 20% of RNA levels were enriched for a variety of AC-rich motifs. For each motif, the cumulative distribution function of the distances from the 5' cap are plotted and compared to a random null distribution of cap distances (see methods). P-values are shown for K-S tests. **B**. Same as "A" for motifs found when AUG containing TLs were removed from the analysis. **C.** Sequences enriched in transcript leaders from the bottom 20% of RNA levels included "AUG" motif with a strong Kozak context and several UC-rich motifs. **D.** Same as "C", for motifs found when excluding TLs with uAUGs.

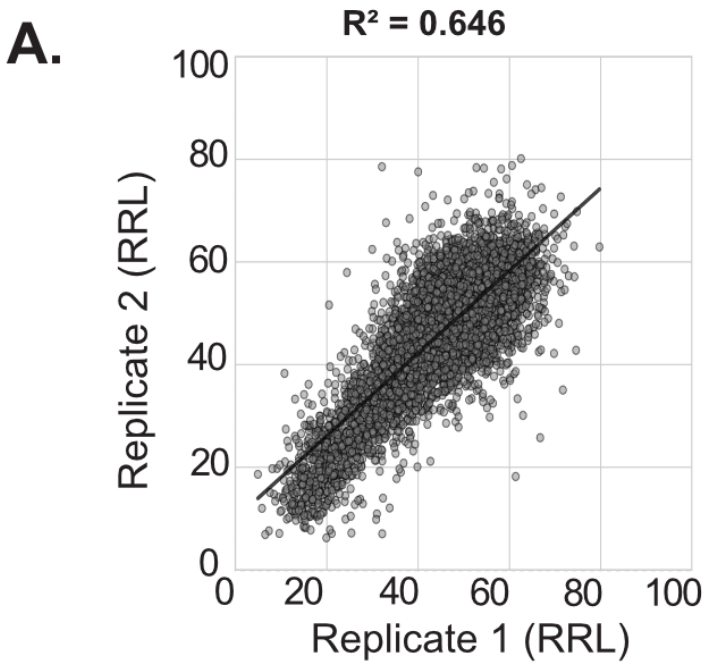


FIGURE S3 | Reproducibility of PoLib-Seq measurements of ribosome loading levels for 5’ TLs. The scatter plots show the comparison of two replicates with the R^2 value listed above the plot.

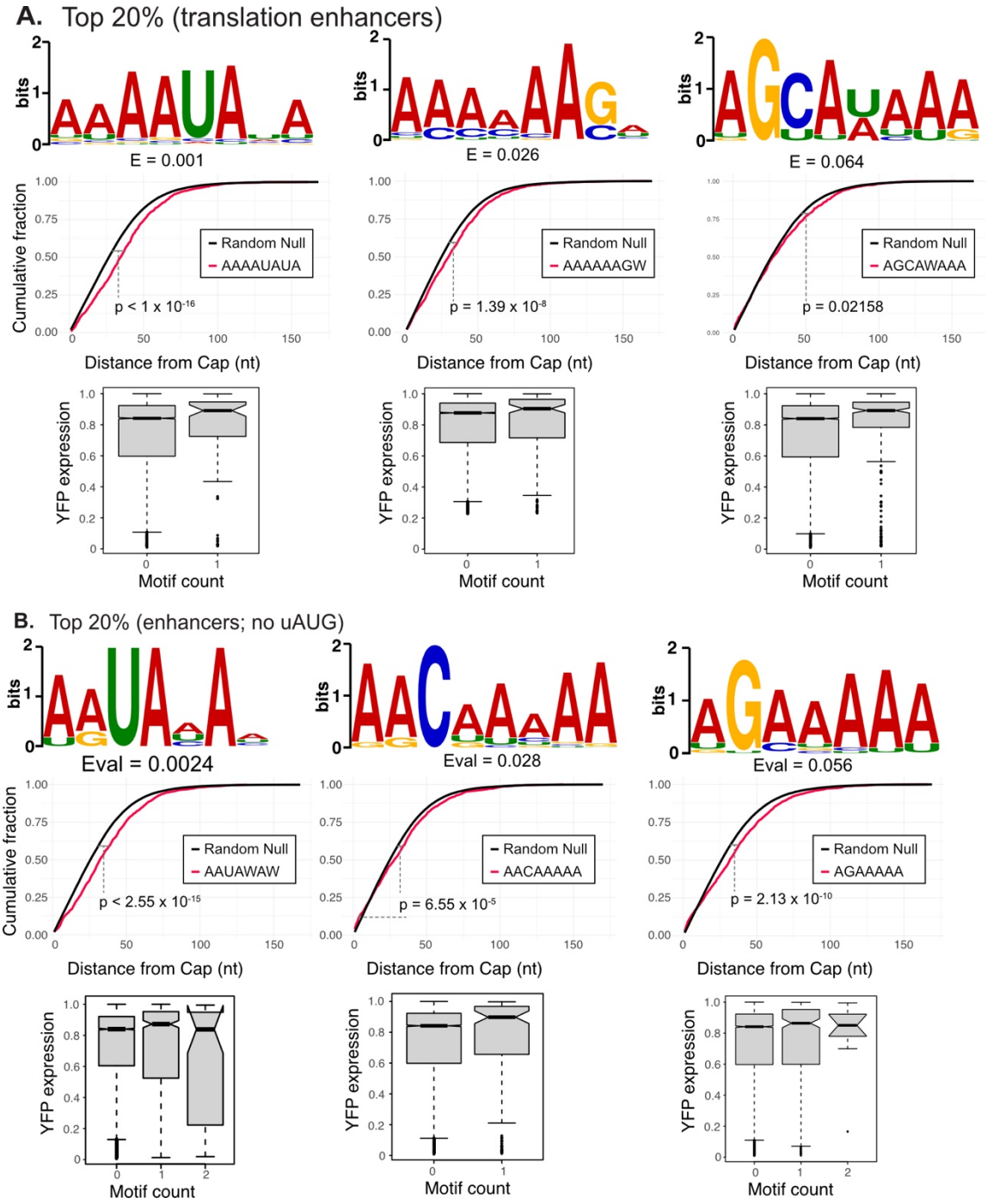


FIGURE S4 | Positional biases of motifs associated with high translation efficiency. Motifs were identified using the STREME algorithm from the MEME suite (c.f. Figure 2). **A.** Sequences in transcript leaders from the top 20% of translation efficiency scores were enriched for a variety of A-rich motifs (above). For each motif, the cumulative distribution function of the distances from the 5' cap are plotted and compared to a random null distribution of cap distances (see methods). The AUA containing motif was slightly enriched internally relative to the 5' cap. Boxplots show the YFP levels for TLs containing zero, one, or two occurrences of each motif. **B.** Same as "A" for motifs found when AUG containing TLs were removed from the analysis.

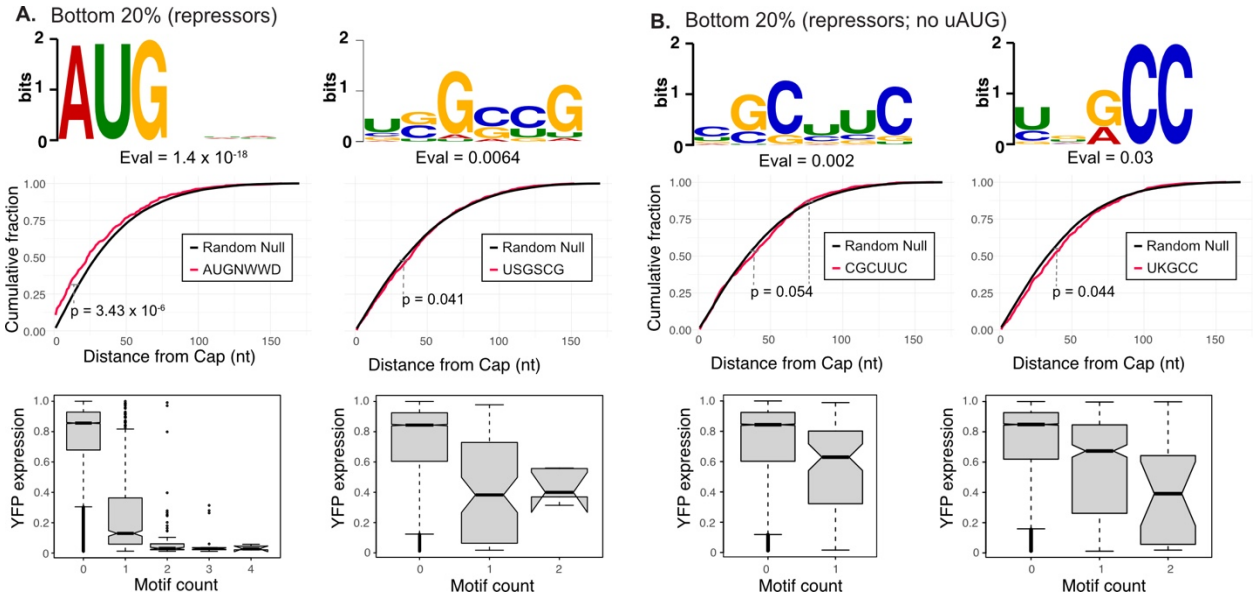


FIGURE S5 | Positional biases of motifs associated with low translation efficiency. Motifs were identified using the STREME algorithm from the MEME suite (c.f. Figure 2). **A.** Sequences in transcript leaders from the bottom 20% of translation efficiency scores were enriched for a variety of A-rich motifs (above). For each motif, the cumulative distribution function of the distances from the 5' cap are plotted and compared to a random null distribution of cap distances (see methods). The upstream AUG motif is enriched near the 5' cap. Boxplots show the YFP levels for TLs containing zero, one, or two occurrences of each motif. **B.** Same as "C" for motifs found when AUG containing TLs were removed from the analysis.

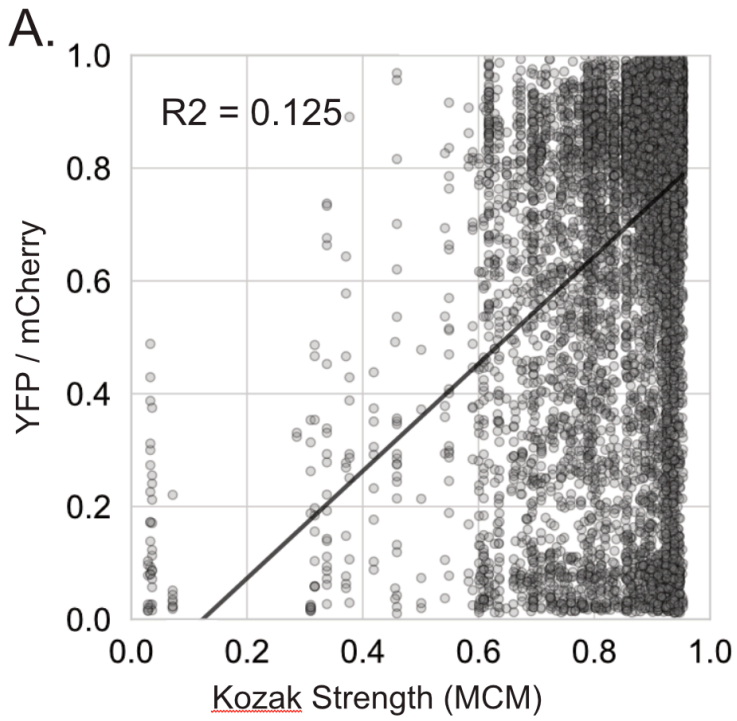


FIGURE S6 | The Main Codon Model (MCM) Kozak strength performs poorly at predicting YFP levels across yeast 5' TLs. This analysis included all TLs, including those with uAUGs and uORFs.

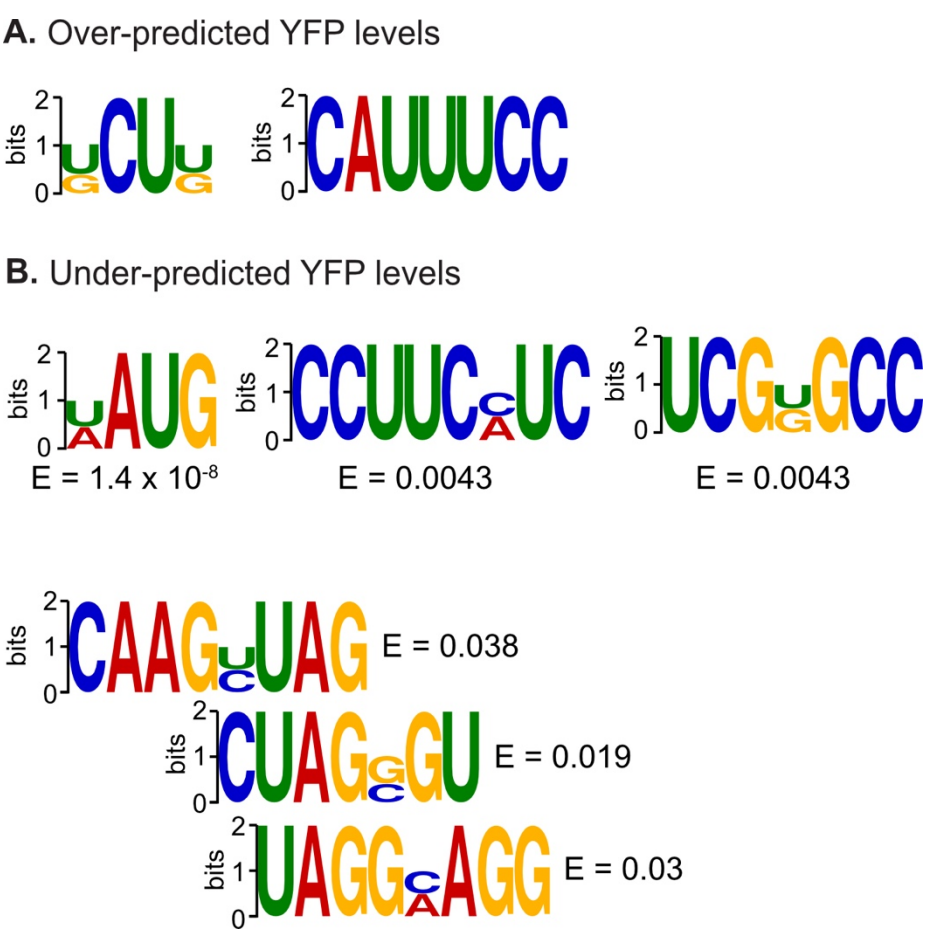


FIGURE S7 | Motifs enriched in over and under-predicted transcript leaders**. A.** Sequences enriched in TLs with expression levels higher than predicted by the HGBR model (overpredicted) were identified using DREME. The [G/U]CU[G/U] motif notably includes "CUG" occurrences, which may function as weak start codons of upstream ORFs. The CAUUUCC motif is very similar to previously defined binding site of SSD1, a known translational inhibitor (Hogan et al., 2009). **B**. Sequences enriched in TLs with expression levels lower than predicted by the HGBR model (underpredicted) included [UA]AUG, which was present in half the underpredicted TLs. Notably, slightly more than half of underpredicted TLs had an AUG that creates an N-terminal extension. As upsteam AUGs are predominantly repressive, by virtue of creating uORFs, the HGBR model predicts that AUGs resulting in N-terminal extension are also repressive, leading to the underprediction of expression. The remaining motifs enriched in underpredicted TLs were rare, occurring in 4 to 5 underpredicted UTRs each.

A. **HG Boost Feature Permutation Importantance (no uAUGs)**

Top 20	Score
LSM Kozak	0.393
Mean RNA	0.080
Distal A Freq.	0.068
Distal C Freq.	0.040
Start Codon (ΔΔG)	0.029
Cap (ΔG)	0.017
CUUC	0.014
Length	0.014
CACC	0.007
Distal G Freq.	0.007
G-quadruplex	0.006
GAUC	0.005
ACCA	0.005
GGGA	0.0048
CAUC	0.0047
AGCC	0.0046
GCUU	0.0044
CAAC	0.0042
GCGG	0.0031
GAGG	0.0030

Additional HG Boost features in supplement

B. **HG Boosting Model (no uAUGs)**

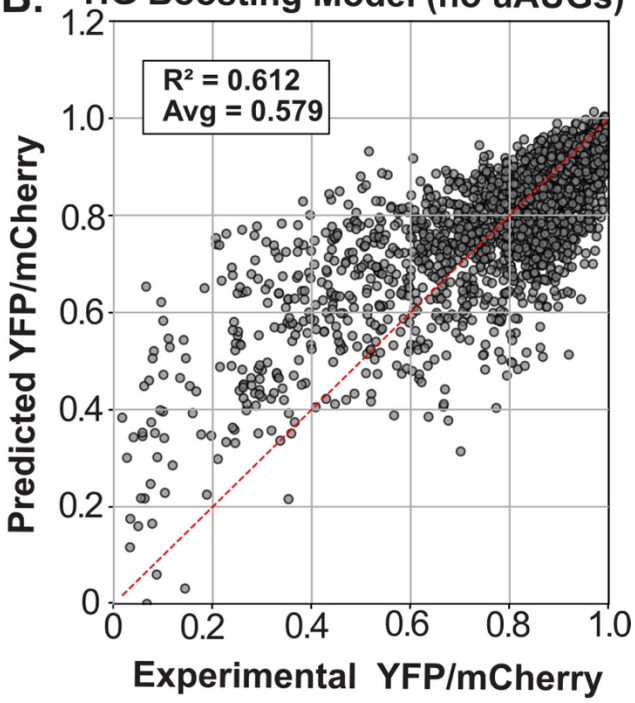


FIGURE S8 | HGBR model of TLs lacking uAUGs (A) Model importance scores for the top 20 most important features extracted from model after n=100 iterations (additional features shown in TABLE S5). (B) The scatter plot shows the measured YFP expression (y-axis) for all 5' TLs versus the HGBR predictions of YFP (x-axis). The resulting model explains ~61b% of variance in experimental YFP.